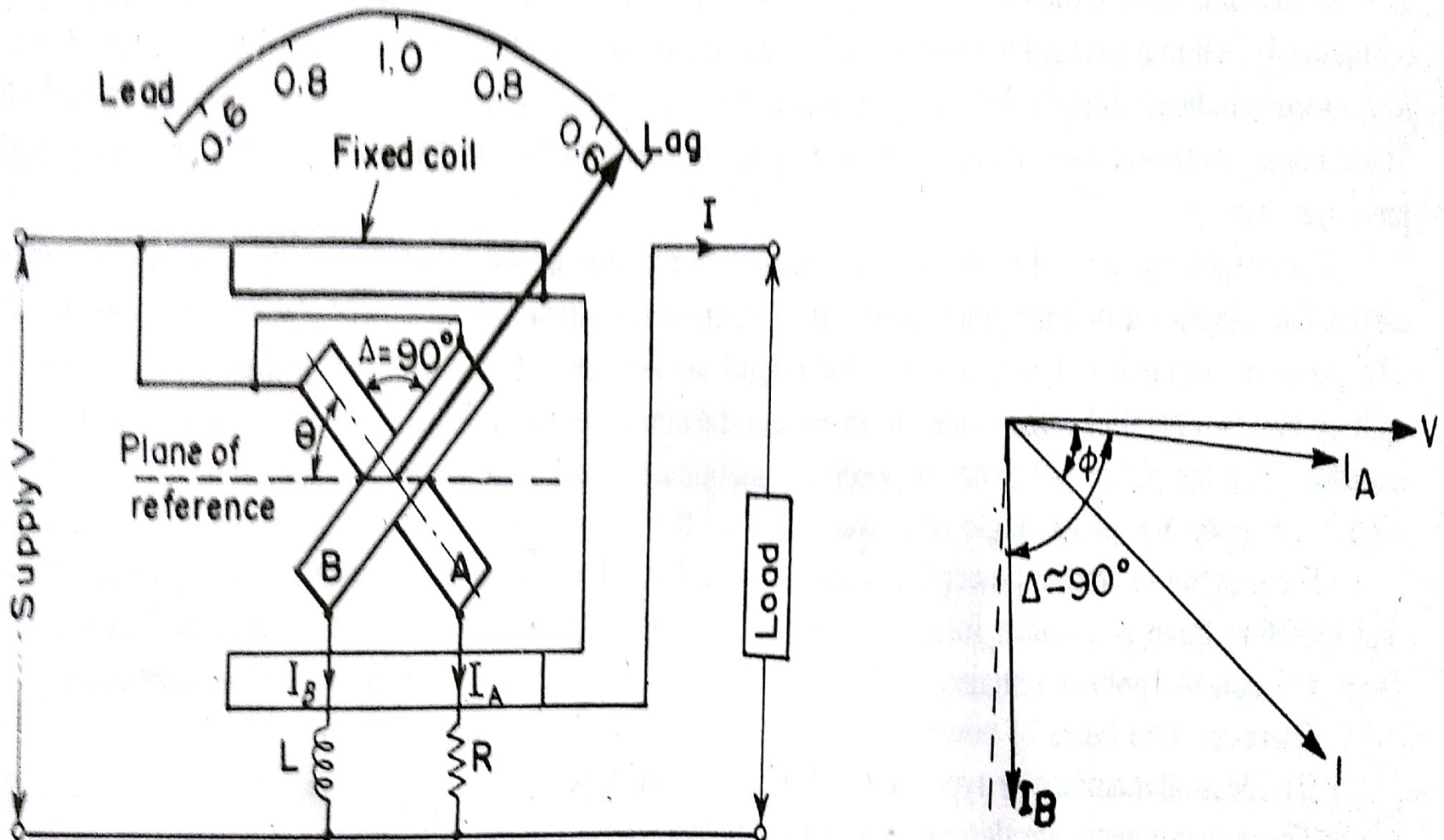


Measurement of Phase and Frequency

P.F. Meter

Single-phase Electrodynamometer type PFM



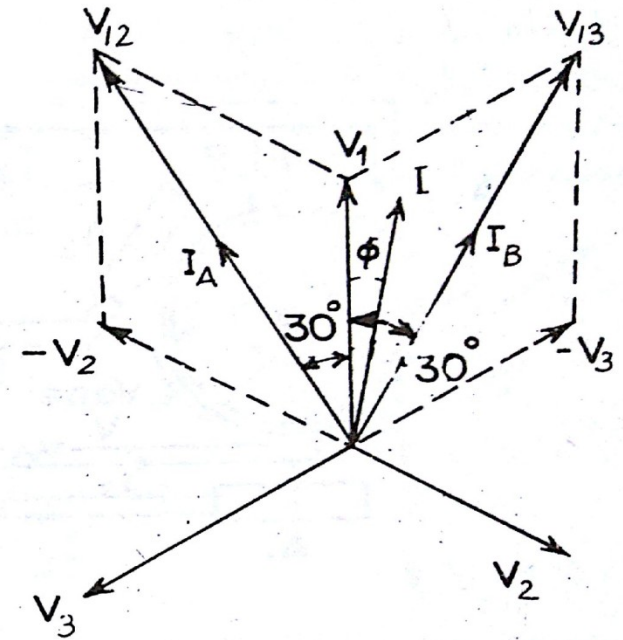
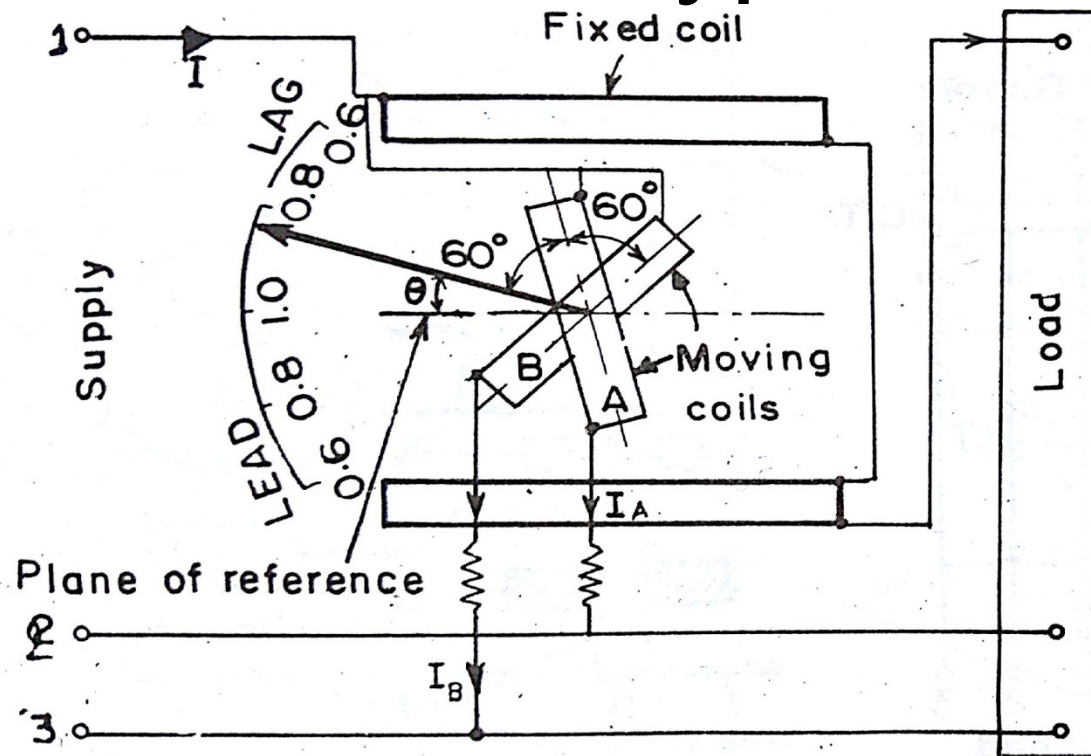
- PC split in two parallel paths - 1 inductive & 1 non-inductive.
- Deflection depends upon phase difference bet. the main current and currents in the two paths of PC.
- Moving system is perfectly balanced by two opposing forces.
- No need of a controlling force.

→ Pointer remains at occupied position.

* Instrument must be designed for, and calibrated at, the f of the supply on which it is to be used.

- * $R = \omega L$
- * I_A & I_B have a difference of 90° .
- * $\Delta = 90^\circ$.
- * Torque on coil A & B are in opposite direction.
- * Pointer takes up a position when the two torques are equal.
- * $T_A = KVI M_{\max} \cos \phi \sin \theta$
- $T_B = KVI M_{\max} \cos (90^\circ - \phi) \sin (90^\circ + \theta)$
i.e. $T_B = KVI M_{\max} \sin \phi \cos \theta$
- * $\theta \rightarrow$ angular deflection from plane of reference
- * At equilibrium $T_A = T_B$
i.e. $\theta = \phi$
- * $M_{\max} \rightarrow$ Max. value of mutual inductance.

Three-phase Dynamometer type PFM



$$T_A = K V_{12} I M_{\max} \cos (30^\circ + \phi) \sin (60^\circ + \theta)$$

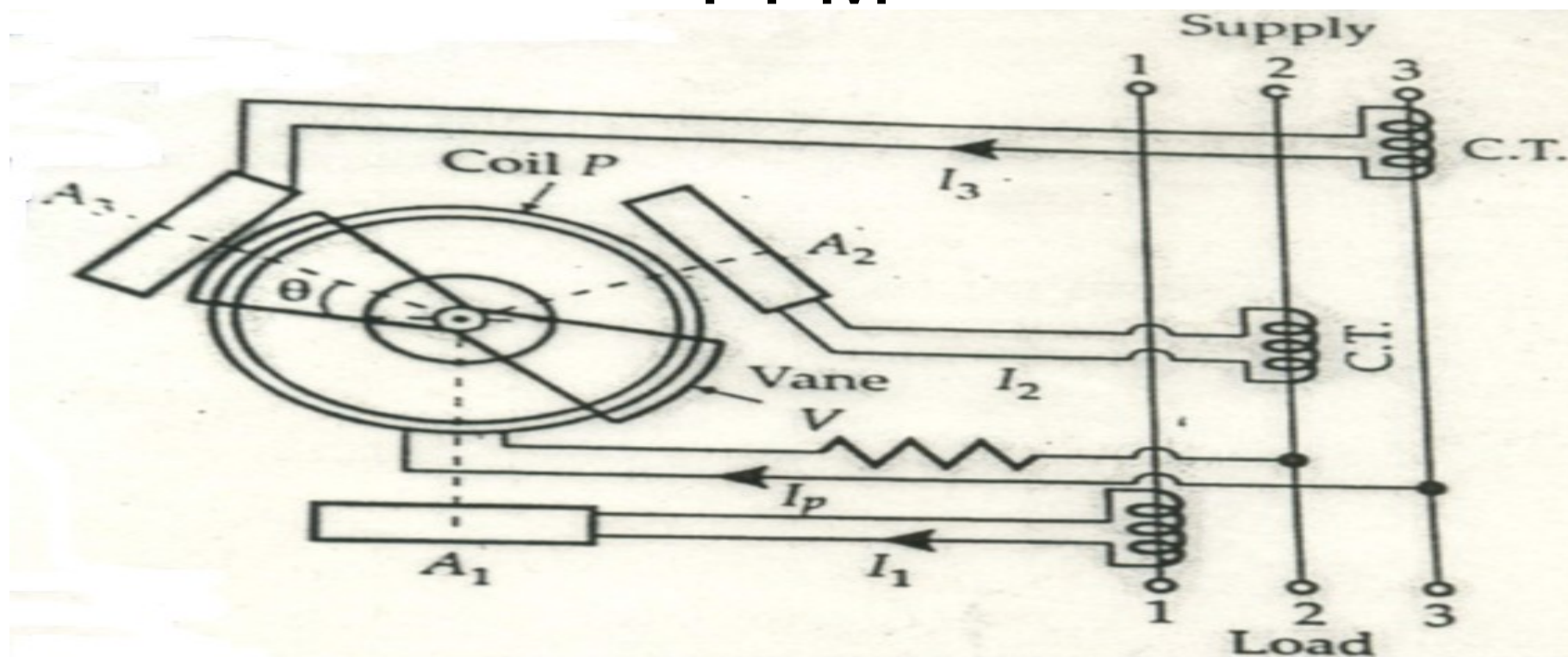
$$= \sqrt{3} K V I M_{\max} \cos (30^\circ + \phi) \sin (60^\circ + \theta)$$

$$T_B = K V_{13} I M_{\max} \cos (30^\circ - \phi) \sin (120^\circ + \theta)$$

$$= \sqrt{3} K V I M_{\max} \cos (30^\circ - \phi) \sin (120^\circ + \theta)$$

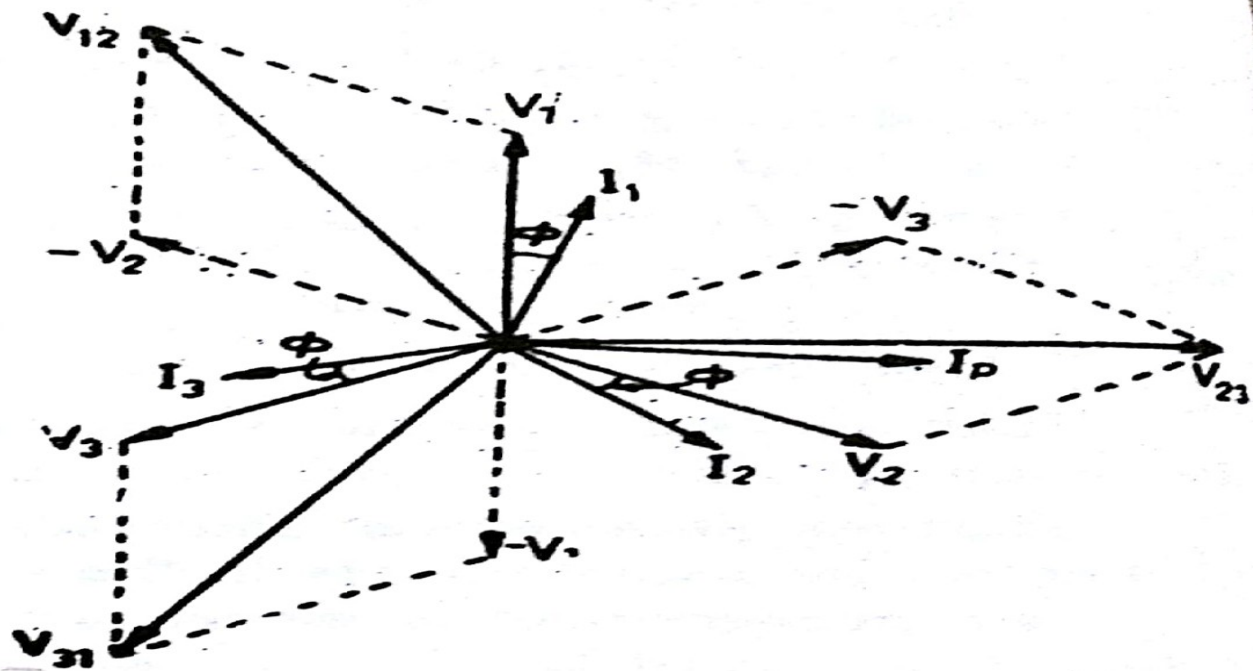
- Only useful for balanced loads.
- Angle between moving coil planes is 120 degree.
- MCs are connected across different phases.
- Phase splitting is not required.
- Indications are independent of frequency of supply.

Rotating Field MI Three-phase PFM



* $A_1, A_2, A_3 \rightarrow$ fixed coils with axes displaced 120° .
Connected to lines 1, 2, 3.

* $P \rightarrow$ fixed coil connected in series with a high resistance across any one pair of lines.



The total deflecting torque

$$T_d \propto [I_1 I_p \cos (90^\circ - \phi) \sin (90^\circ + \theta) + I_2 I_p \cos (330^\circ - \phi) \sin (210^\circ + \theta) + I_3 I_p \cos (210^\circ - \phi) \sin (330^\circ + \theta)].$$

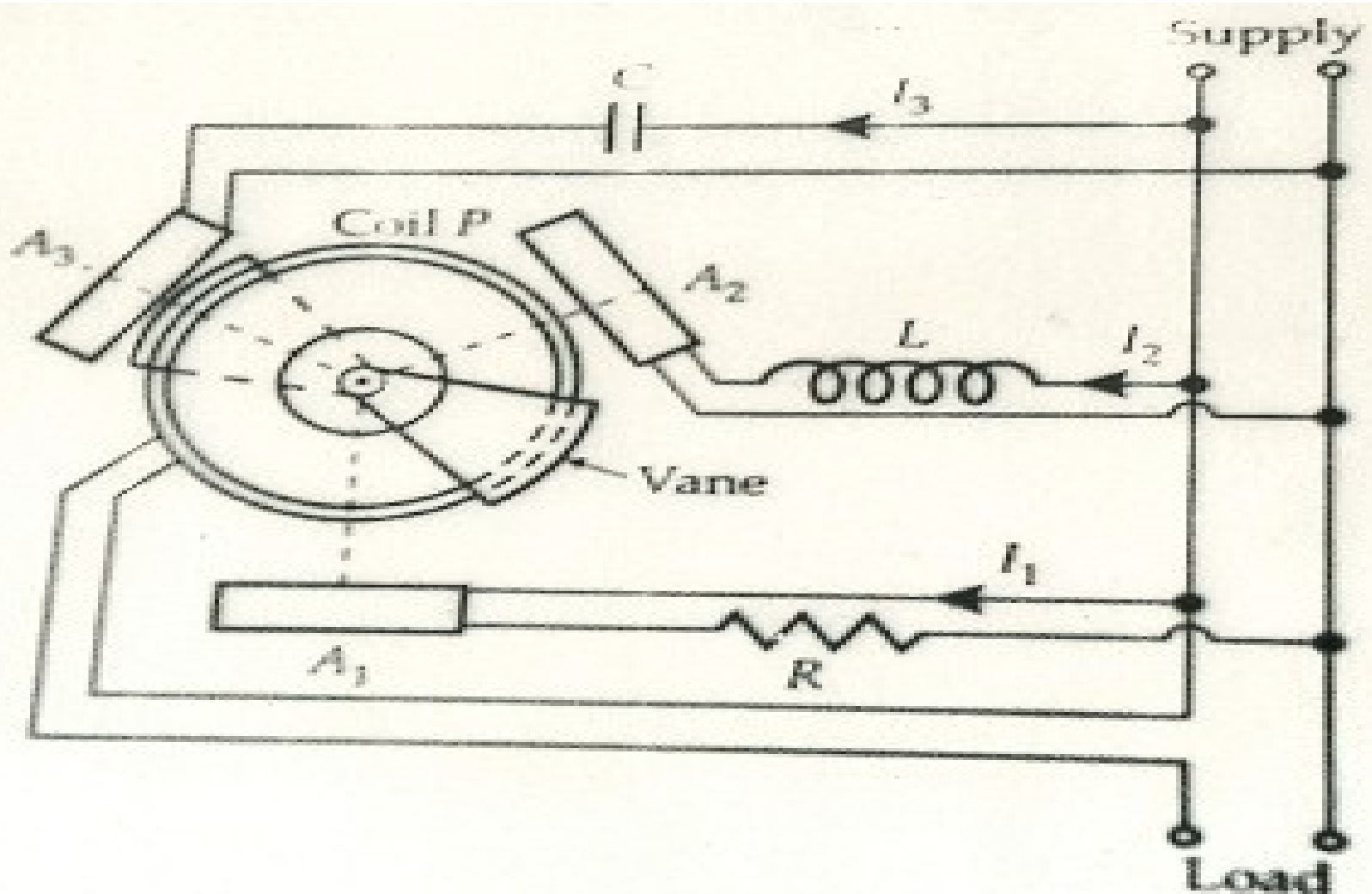
For a steady deflection, the total torque must be zero. Also considering the system to be balanced *i.e.*, $I_1 = I_2 = I_3$, we have,

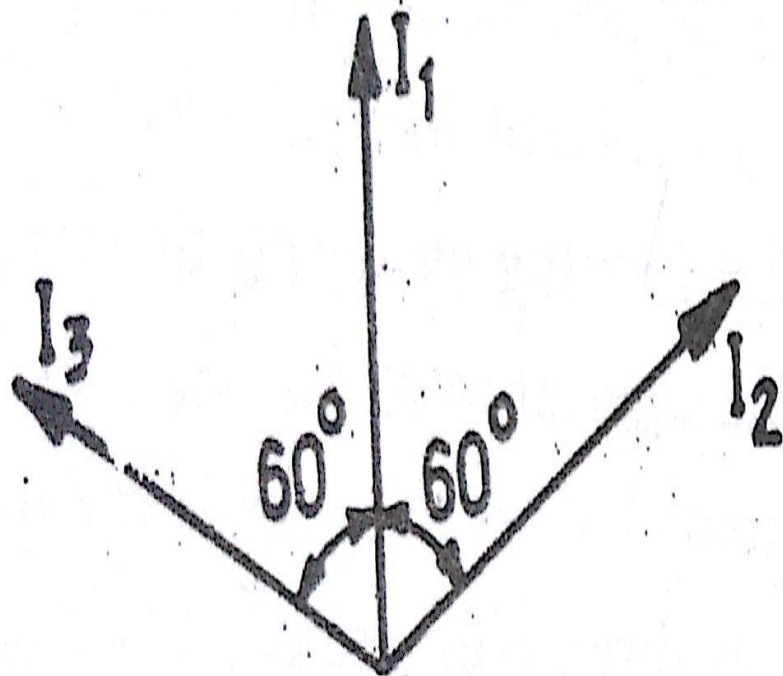
$$\cos (90^\circ - \phi) \sin (90^\circ + \theta) + \cos (330^\circ - \phi) \sin (210^\circ + \theta) + \cos (210^\circ - \phi) \sin (330^\circ + \theta) = 0$$

Solving the above expression, we have : $\theta = \phi$.

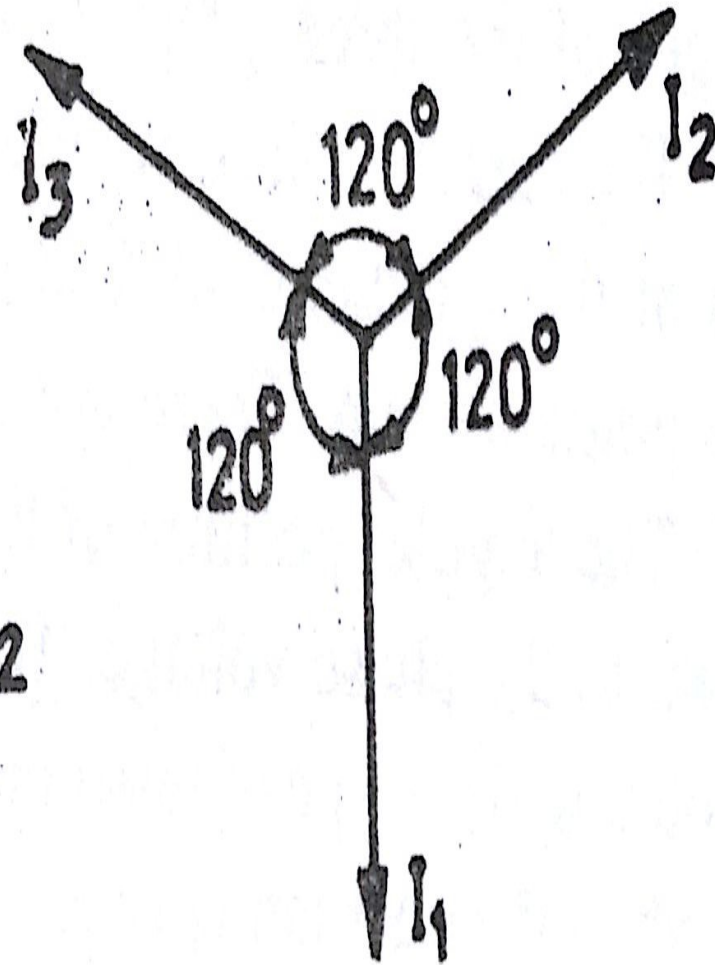
Therefore the deflection of iron vane from the reference axis is a direct measure of the phase angle between each line current and the corresponding phase voltage.

Rotating Field MI Single-phase PFM





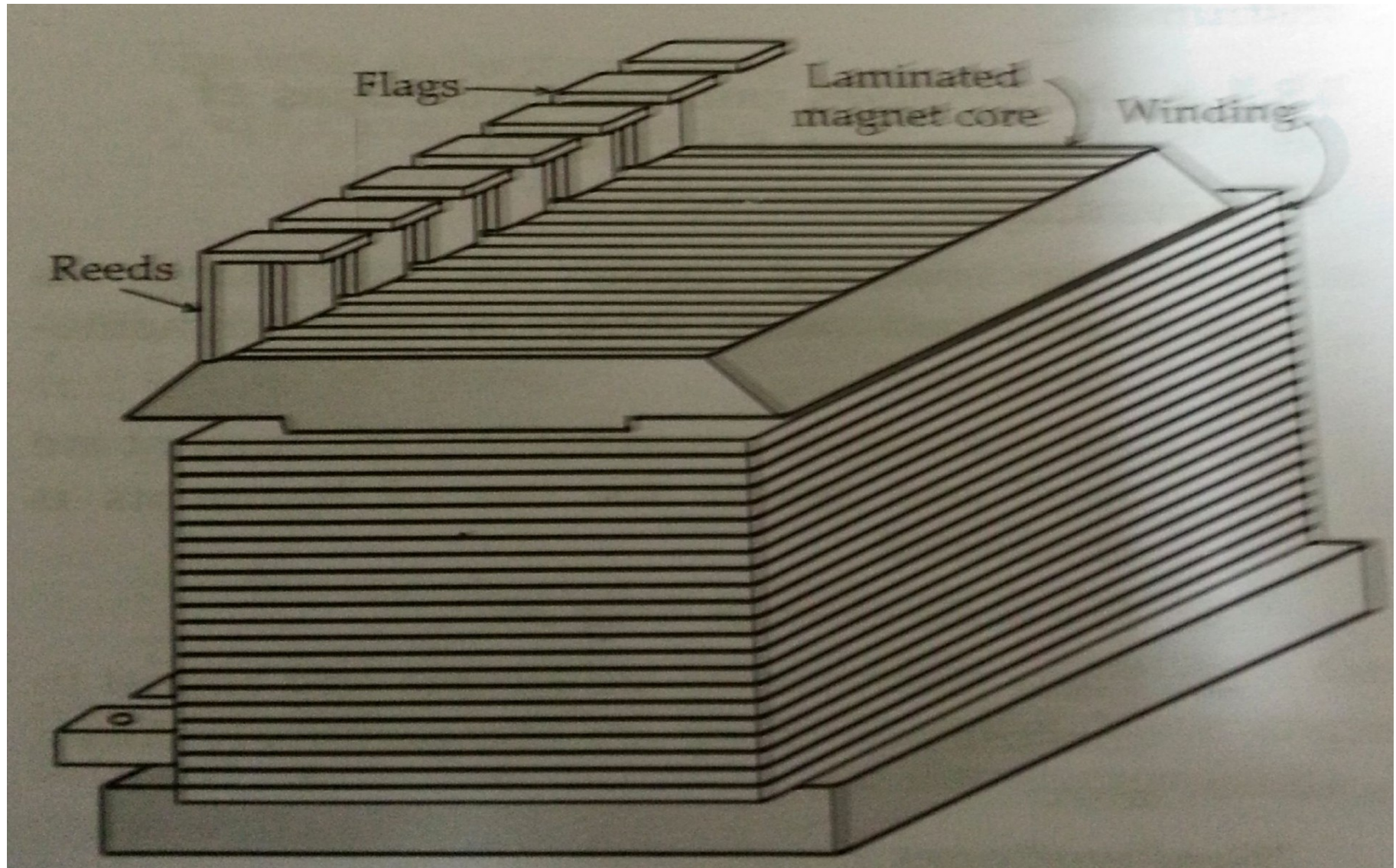
(i) Before reversal of connections of coil A_1



(ii) After reversal of connections of coil A_1

Frequency Meter

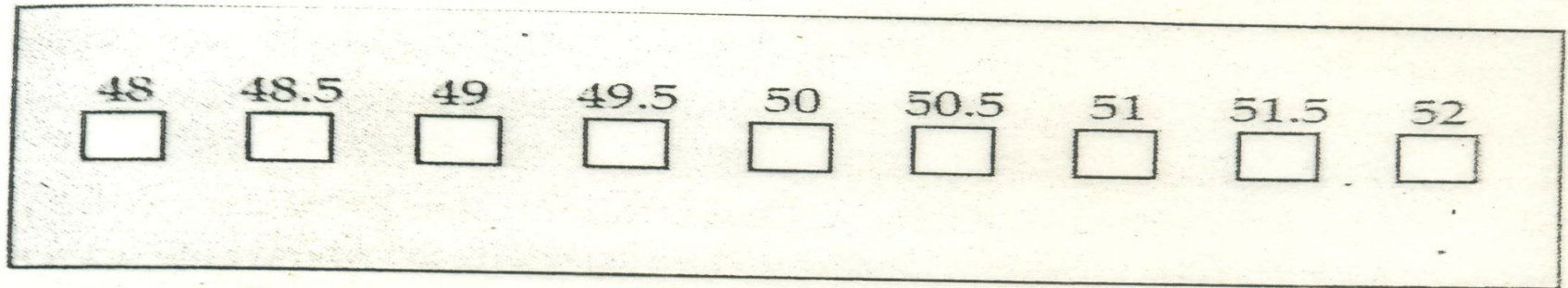
Vibrating Reed Type



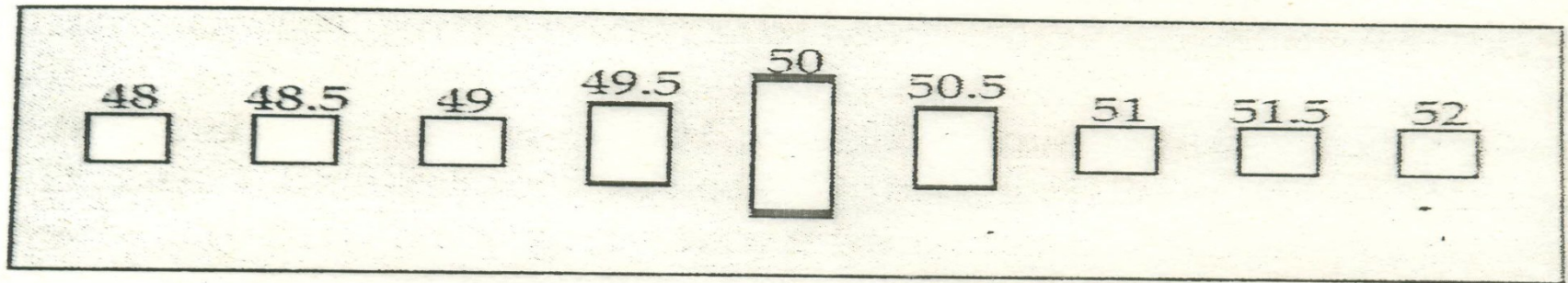
VIBRATING REED TYPE FREQUENCY METER

- Reeds are thin steel strips.
- Reeds are placed alongside an electromagnet.
- Electromagnet is connected across supply whose f is to be measured.
- Natural f of vibration of the reeds are different.
- Reading can be taken by resonating reed.

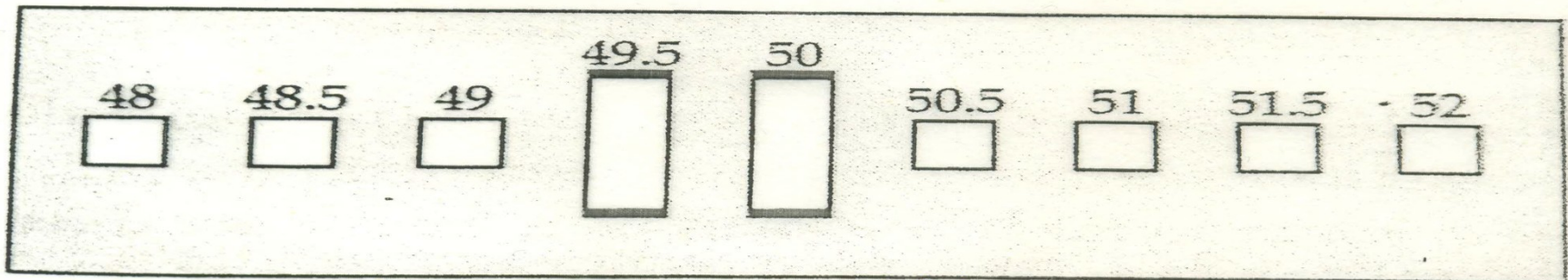
Indication from Vibrating Reeds



(a)

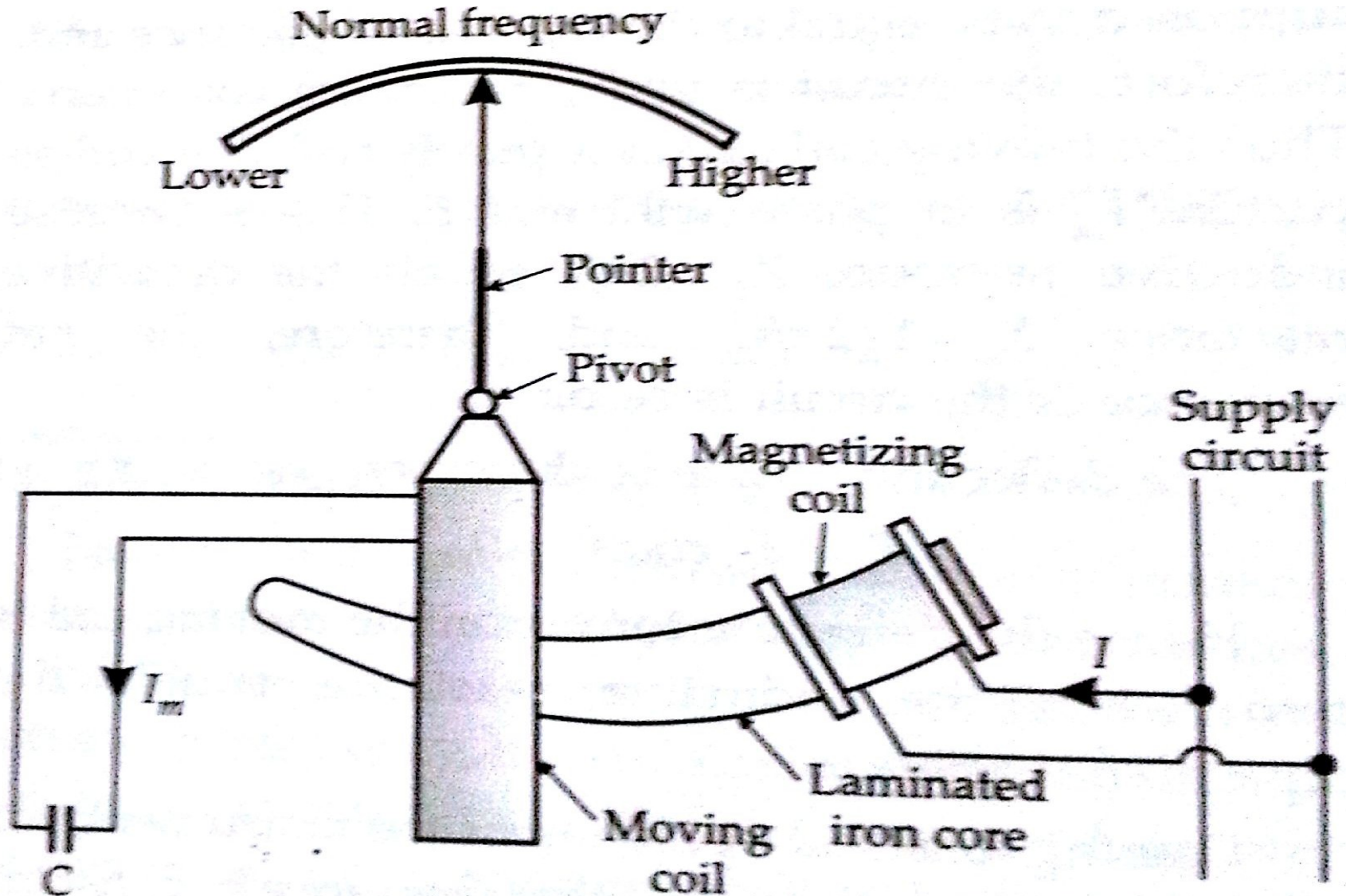


(b)



(c) 'f' bet. 49.5 & 50 Hz

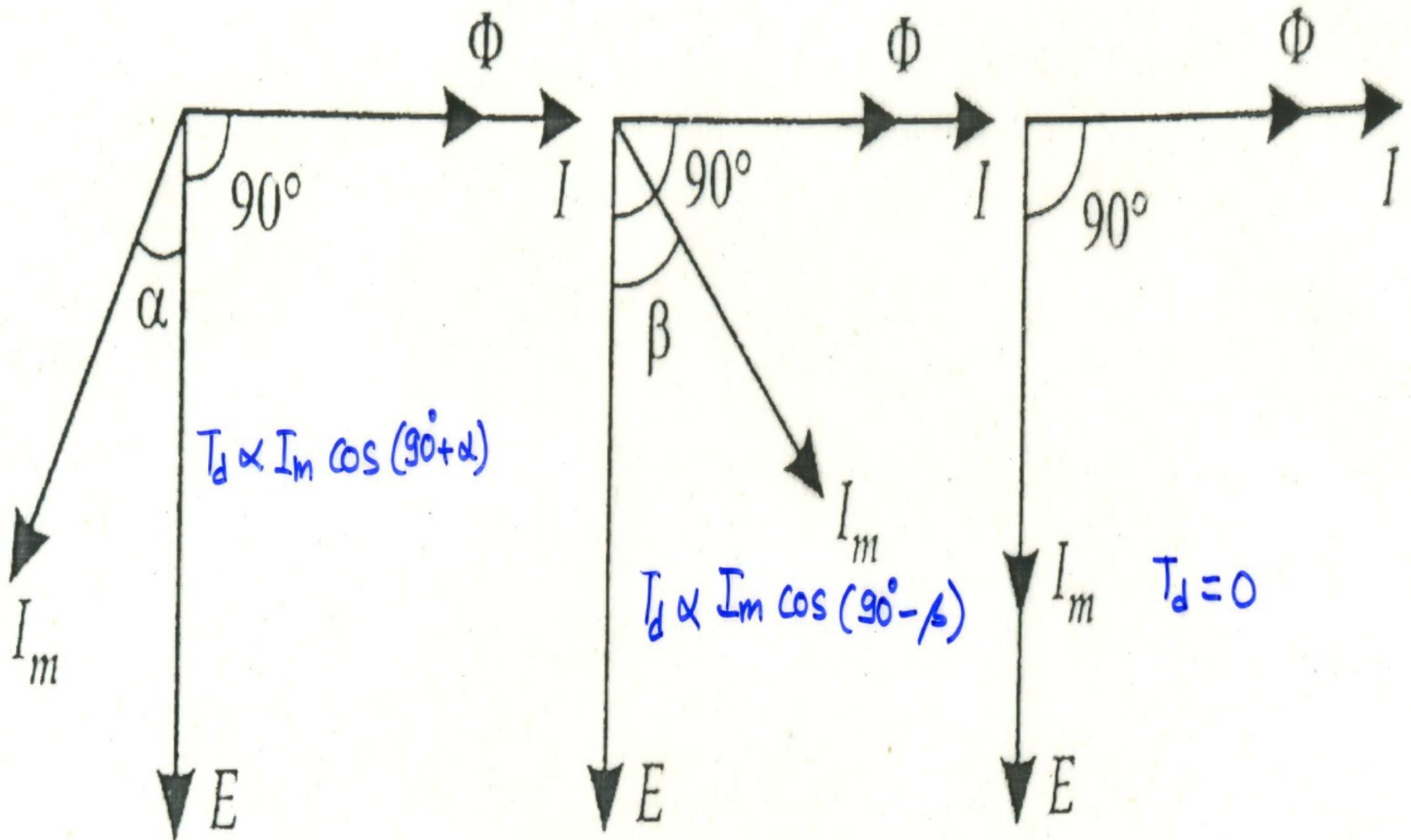
Electrical Resonance Type



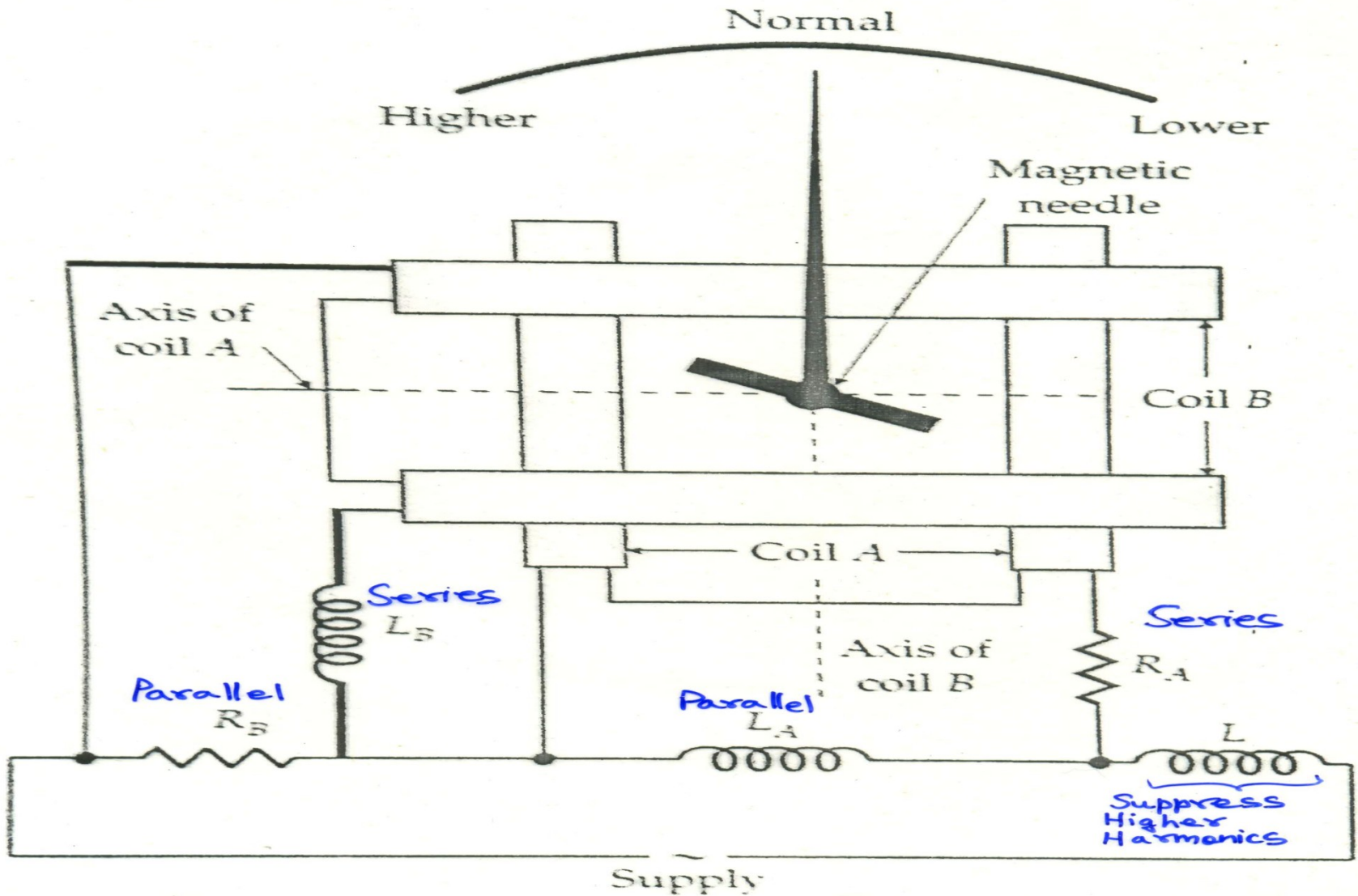
FERRODYNAMIC TYPE FREQUENCY METER

- A fixed coil is connected across supply whose f is to be measured.
- No provision for a controlling force.
- For a fixed f X_c is constant but X_L varies. Inductance of MC is dependent on the position it occupies on iron core. Inductance is max. when moving coil is close to magnetizing coil.

Phasor Diagrams



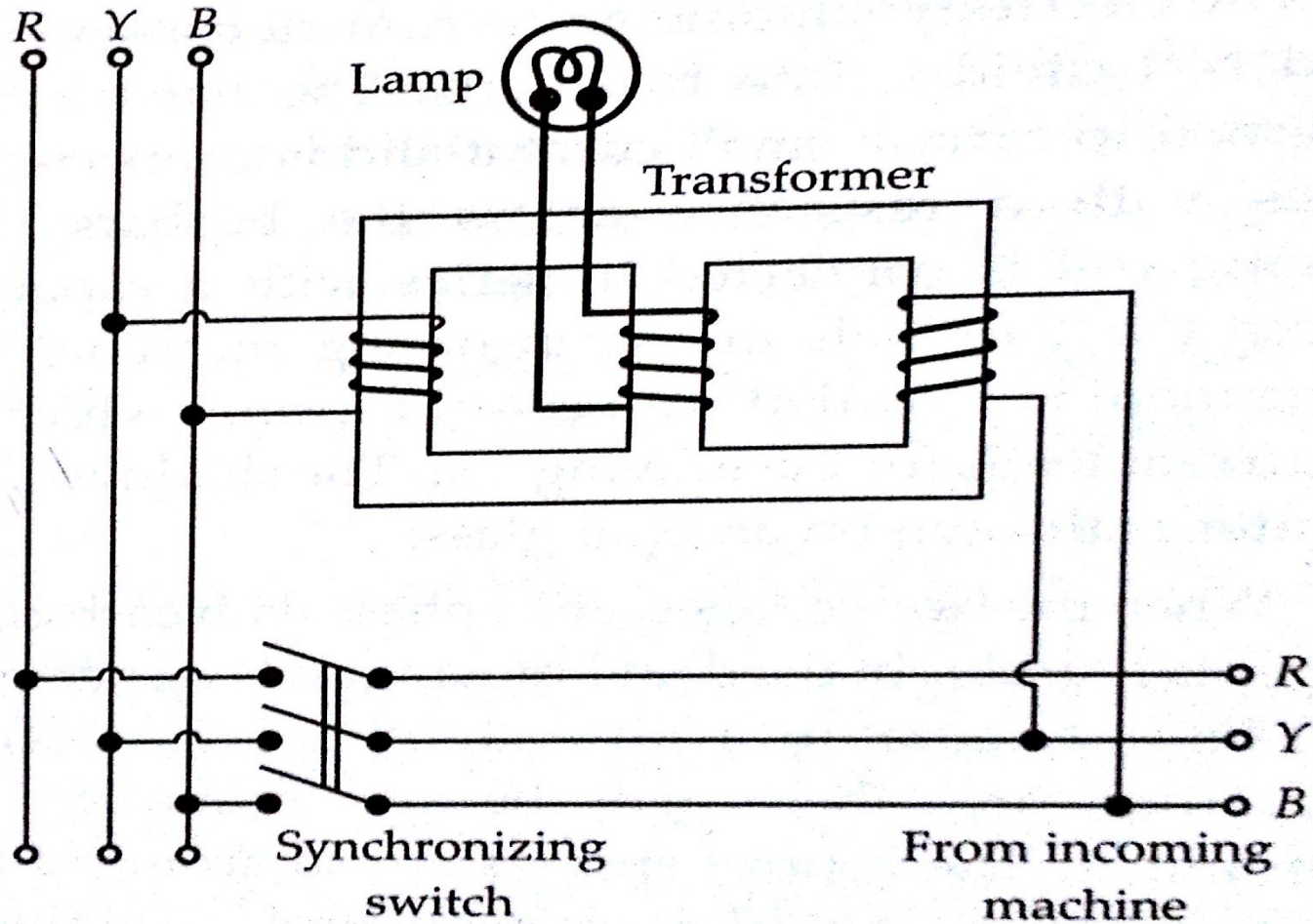
Weston FM



Synchroscope

- Determines the correct instant for closing the switch which connects an alternator to power station busbars.
- Correct instant of synchronizing is when busbar and incoming alternator voltages are equal in magnitude, phase and frequency.

Weston Type Synchroscope



- Consists of a three limbed transformer.
- One outer limb is excited from busbars while the other is excited by the incoming machine.
- Central limb winding is connected to a lamp.
- Resultant flux through the central limb is the phasor sum of the outer limb fluxes.
- The correct instant of synchronizing is when the lamp flickers at a very slow rate and is at its maximum brightness.
- Flickering is due to the difference in frequencies of the busbar and the incoming machine.